UBT802: NANOBIOTECHNOLOGY

L	T	P	Cr
3	1	0	3.5

Course Objective: This course introduces the concepts of quantum confinement, synthesis routes and characterization tools for nanomaterials to the students. It also provides preliminary ideas about choice of materials for biomedical, therapeutic and environmental applications.

Syllabus

Introduction to Nanoscience: Features of nanosystems, ccharacteristic length scales of materials and their influence on properties, quantum size effect: electron confinement in 2D, 1D and 0D, quantum nanostructures.

Synthesis and Characterization of Nanomaterials: Bottom-up and top-down approaches, thin film deposition techniques, biosynthesis of nanoparticles and self-assembly, **Characterization of nanomaterials:** X-ray diffraction, electron microscopy (SEM and TEM), atomic force microscopy and spectroscopy techniques (UV-visible, fluorescence, FTIR and RAMAN spectroscopy), Thermal characterization techniques (TGA, DTA, DSC).

Biomedical Applications: Concepts and working principles of Targeted drug delivery systems, photodynamic therapy, magnetic hyperthermia, nano-antimicrobials, nanobiosensors, etc.

Nanotoxicology: Cytotoxic and genotoxic effects of nanomaterials, toxic effects on environment (plants and microbes), impact of nanotechnology on society and industry.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)
The students will be able to:

1. Explain the effects of quantum confinement on physical, chemical and biological properties of materials at nanoscale
2. Choose an appropriate synthesis technique to synthesize nanostructures of desired size, shape and surface properties
3. Choose appropriate nanostructures for biomedical applications
4. Evaluate the potential toxic effects of nanotechnology on living organisms

Text Books

1. “Introduction to Nanotechnology” Poole,C.P., Owens,F.J., John Wiley &Sons (2003).
2. "Nanostructures and Nanomaterials: Synthesis, Properties and Applications", G. Cao, Imperial College Press (2004).
3. Nanobiotechnology; Concepts, Applications and Perspectives", C. M. Niemeyer, C. A. Mirkin, Wiley-VCH (2004).
4. "Bionanotechnology: In Nanoscale Science and Technology", G. J. Leggett, R. A. L. Jones, John Wiley & Sons, (2005).
5. "Nano: The Essentials", T. Pradeep, Tata McGraw-Hill Publishing Company Ltd. (2007).

Reference Books

1. "Bionanotechnology", D. S. Goodsell, John Wiley & Sons (2004).
2. "Springer Handbook of Nanotechnology", Eds: Bhushan, 2nd edition.
3. "Encyclopedia of Nanoscience and Nanotechnology", Eds: H. S. Nalwa, American Scientific Publishers (2004).

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessional (May include assignments/tutorials/quizzes)	30